

What is this?

*This notebook is part of a collection of *pLuto* notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.*

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TIME-DOMAIN ASTROPHYSICS

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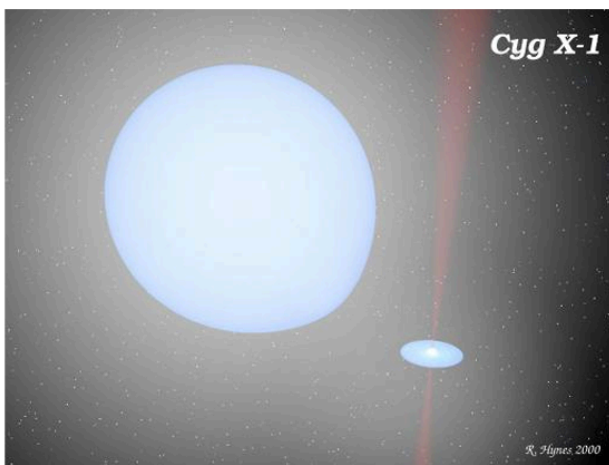


Science Case: X-Ray Binaries

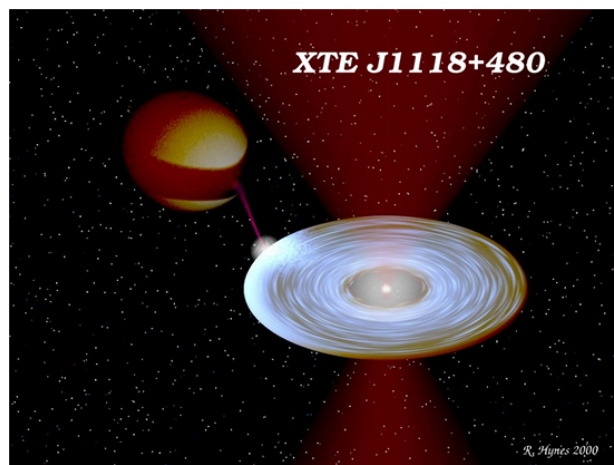
X-ray Binaries (XRBs)

- X-ray binaries are a class of binary stars that are luminous in X-rays.
- The X-rays are produced by matter falling from one component, called the “donor” (usually a relatively normal star), to the other component, called the “accretor”.
- The latter is very compact: a neutron star or a black hole.
- The infalling matter releases gravitational potential energy, up to several tenths of its rest mass, as X-rays.
 - $L_X \sim 10^{35-39} \text{ erg s}^{-1}$ in the X-ray band.
- The donor star nature affects the time evolution of the binary.

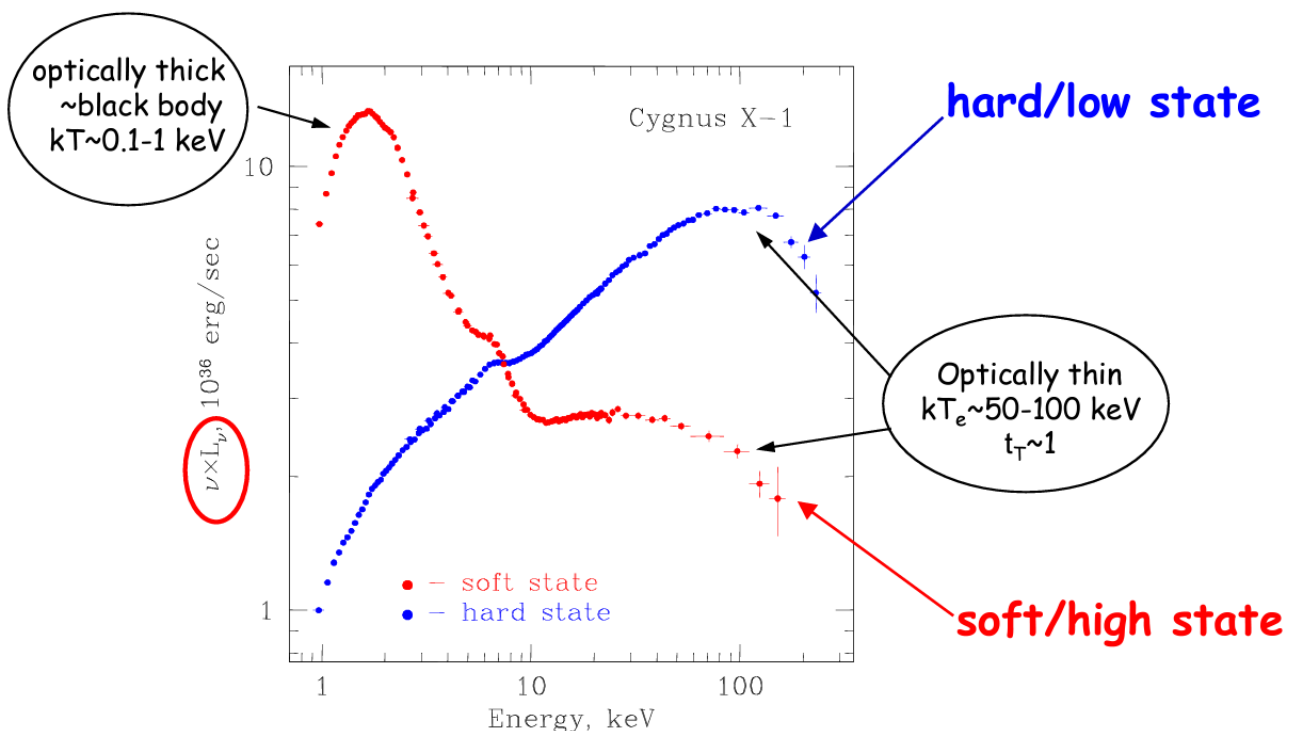
High mass



Low mass

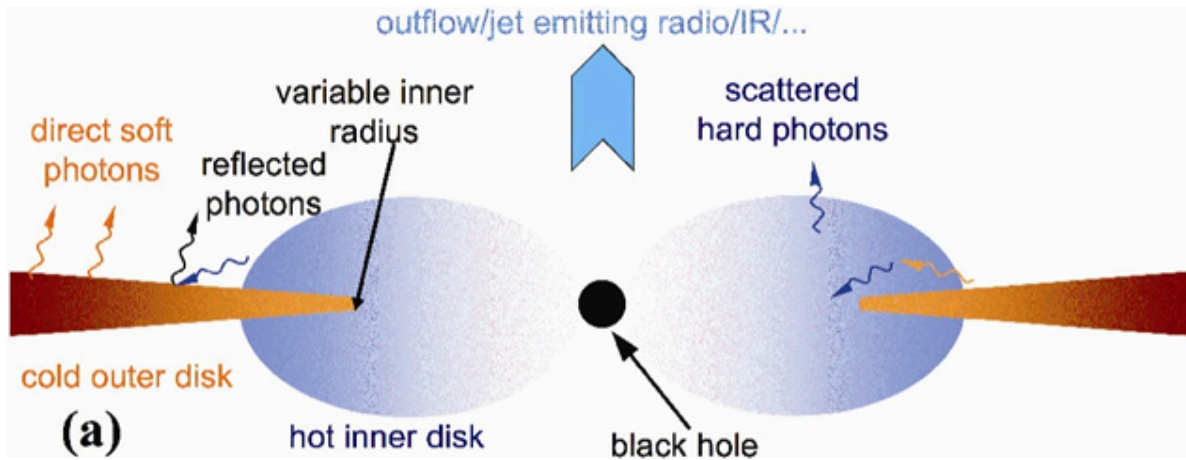


- Emission can be observed under different “states”.



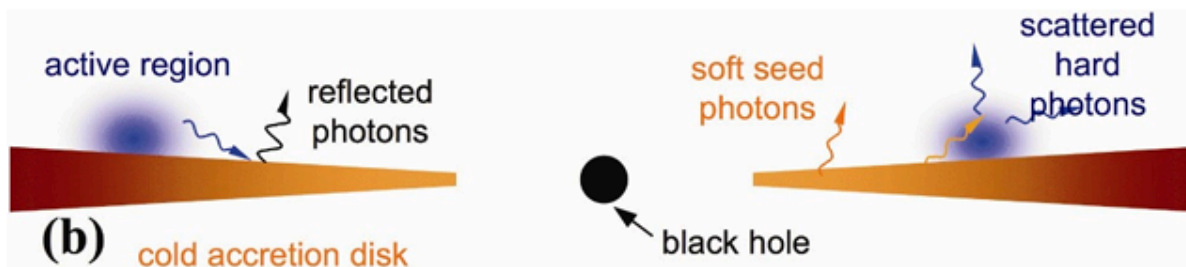
hard state

truncated disk + inner hot flow



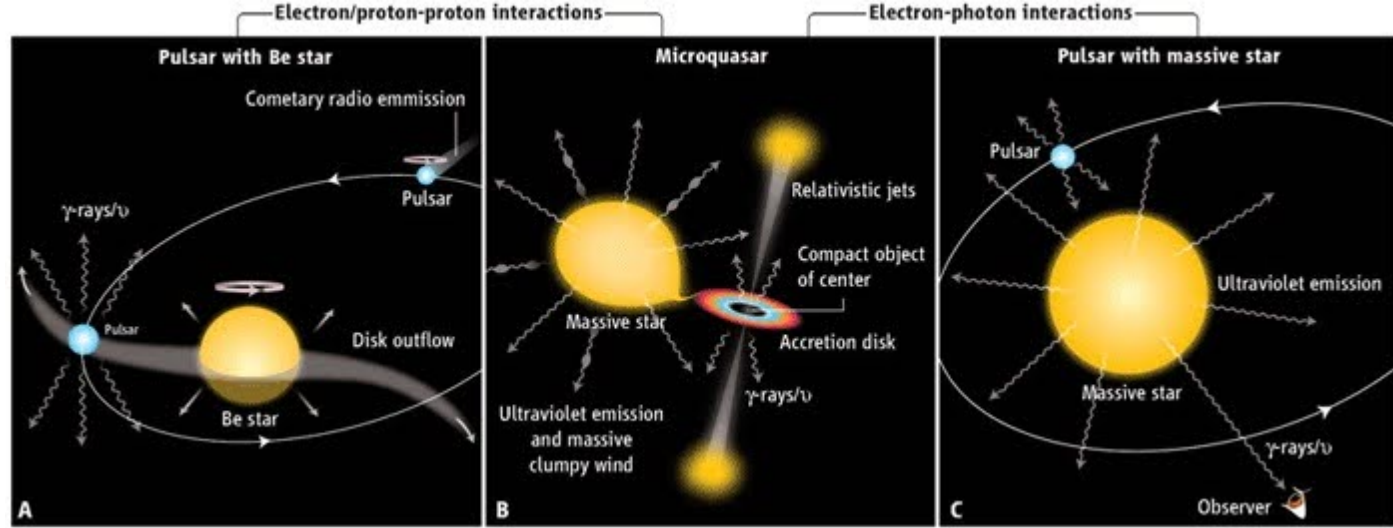
soft state

disk extends to last stable orbit



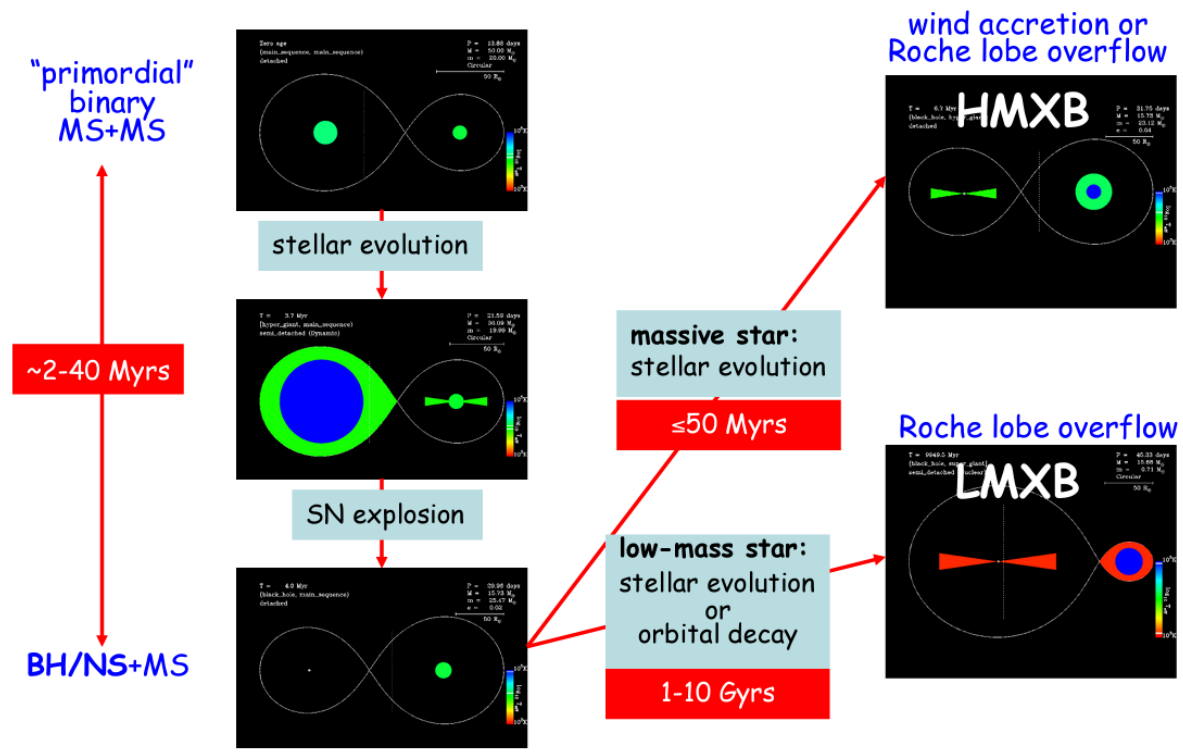
from Zdziarski & Gierlinski, 2004

- The lifetime and the mass-transfer rate in an X-ray binary depends on the evolutionary status of the donor star, the mass ratio between the stellar components, and their orbital separation.
- X-ray binaries are further subdivided into several (sometimes overlapping) subclasses. Note that the classification by mass (high, intermediate, low) refers to the optically visible donor, not to the compact X-ray emitting accretor.



XRB evolution

- This is a very simple sketch of binary evolution



- Evolutionary time scales:
 - XMXB: $t \sim 10 - 50$ Myrs, comparable to duration of star forming event so that these systems are star formation tracers.
 - LMXB: $t \sim 1 - 10$ Gyrs, comparable to live time of the host galaxies so that these systems are stellar mass tracers.

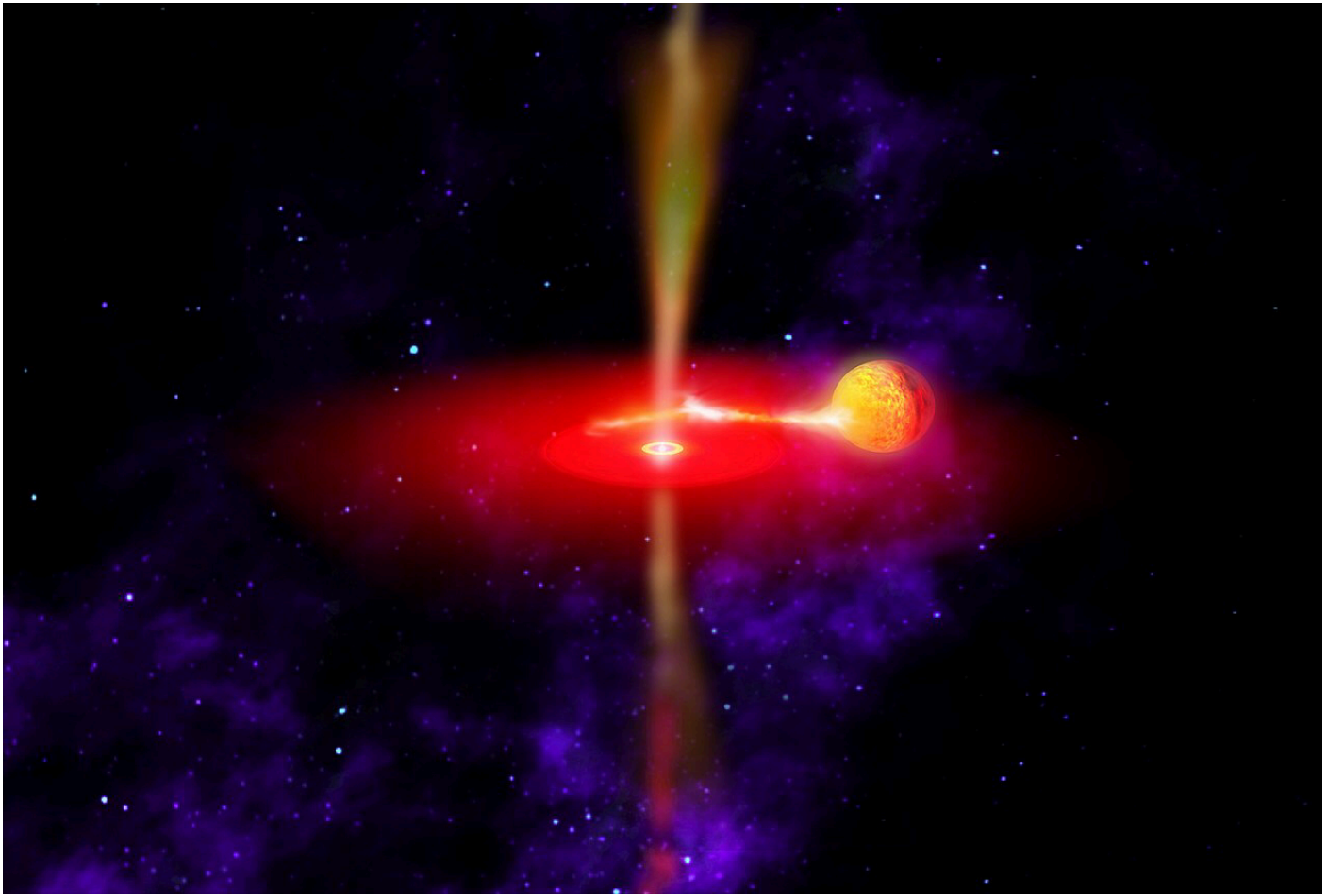
XRB Zoo

- Low-mass X-ray binaries (LMXBs)
 - Soft X-ray transients (SXTs)
 - Symbiotic X-ray binaries
 - Super soft X-ray sources or Super soft sources (SSXs), (SSXB)
- Intermediate-mass X-ray binaries (IMXBs)
 - Ultracompact X-ray binaries (UCXBs)
- High-mass X-ray binaries (HMXBs)
 - Be/X-ray binaries (BeXRBs)
 - Supergiant X-ray binaries (SGXBs)
 - Supergiant Fast X-ray Transients (SFXTs)
- Others
 - X-ray bursters
 - X-ray pulsars
 - Microquasars (radio-jet X-ray binaries that can house either a neutron star or a black hole)

GX339-4 / V821 Ara

.....

- GX-339-4 is a Galactic Low Mass X-ray Binary (LMXB), and candidate black-hole.
- It is a variable source showing occasionally a flaring activity.
- During the outbursts GX 339-4 shows evolution of quasi-periodic oscillations (QPOs).
- A strong, variable relativistic jet, emitting from radio to infrared wavelengths was observed by several studies.
- An artistic view of the GX339-4 system:



Exercise: the QPO in GX339

- We are going to analyse an observation of this source carried out by the Rossi X-ray Timing Explorer (RXTE) satellite.

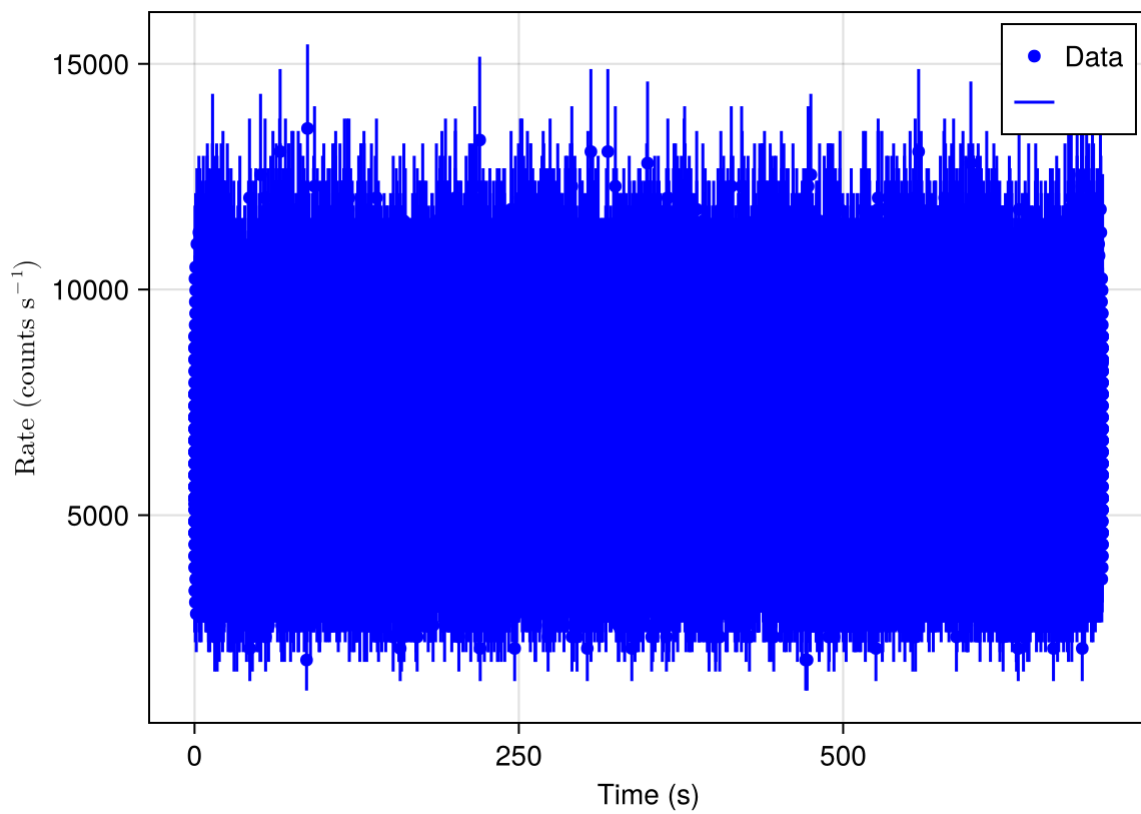
	TIME	RATE	ERROR	FRACEXP
1	5.14597e8	5250.17	1237.48	0.877686
2	5.14597e8	7680.0	1402.17	1.0
3	5.14597e8	6656.0	1305.35	1.0
4	5.14597e8	6656.0	1305.35	1.0
5	5.14597e8	5888.0	1227.73	1.0
6	5.14597e8	7424.0	1378.6	1.0
7	5.14597e8	8960.0	1514.52	1.0
8	5.14597e8	4096.0	1024.0	1.0
9	5.14597e8	6144.0	1254.14	1.0
10	5.14597e8	8704.0	1492.72	1.0
altro				
179201	5.14598e8	8371.86	4185.93	0.122314

- **FRACEXP** is the fraction of a bin of length Δt effectively covered by the observations. We don't need it, but this information can be useful for a careful estimate of the statistics of the light curve.
- Let's look at the spacing of the observations.

Sampling, mean: 0.00391s, variance: 0.00000s

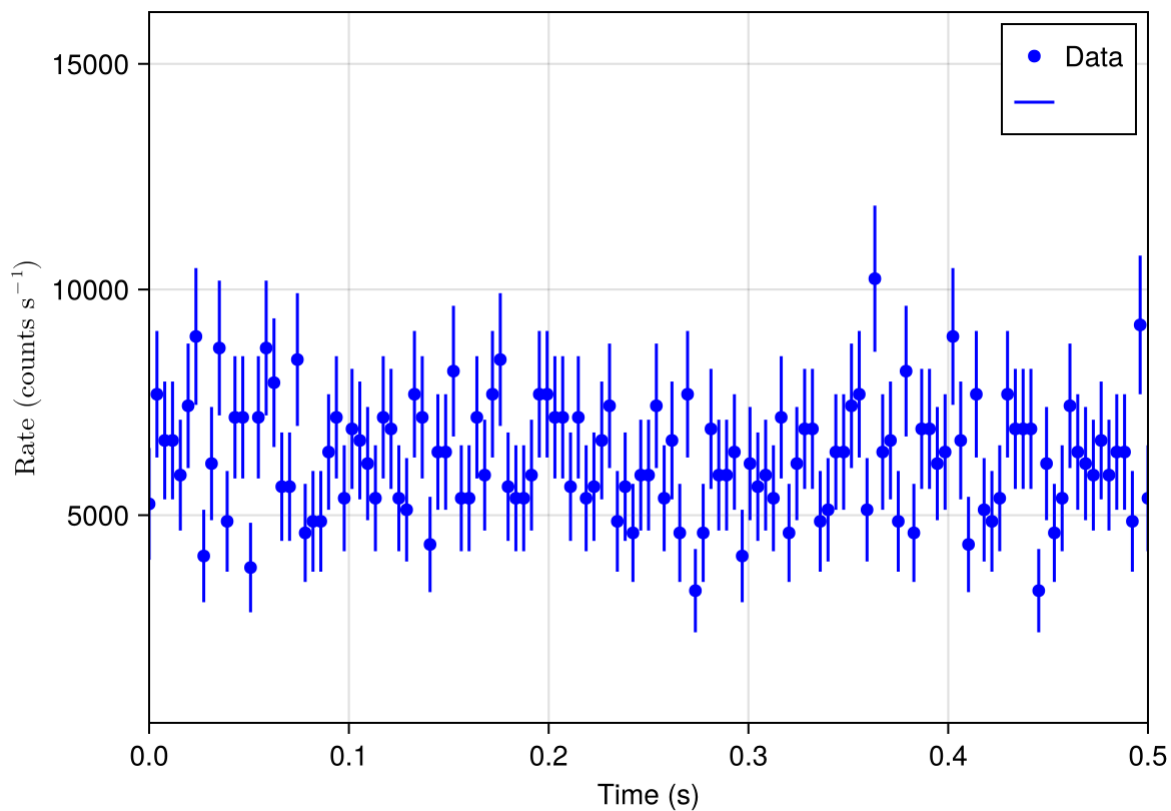
- Variance "0" means that all time bins are of the same length and the sampling time turns out to be 3.9ms. Therefore the Nyquist frequency is about 128Hz.
- Let's rename, rescale and plot our input variables:

RXTE observation of GX339-4

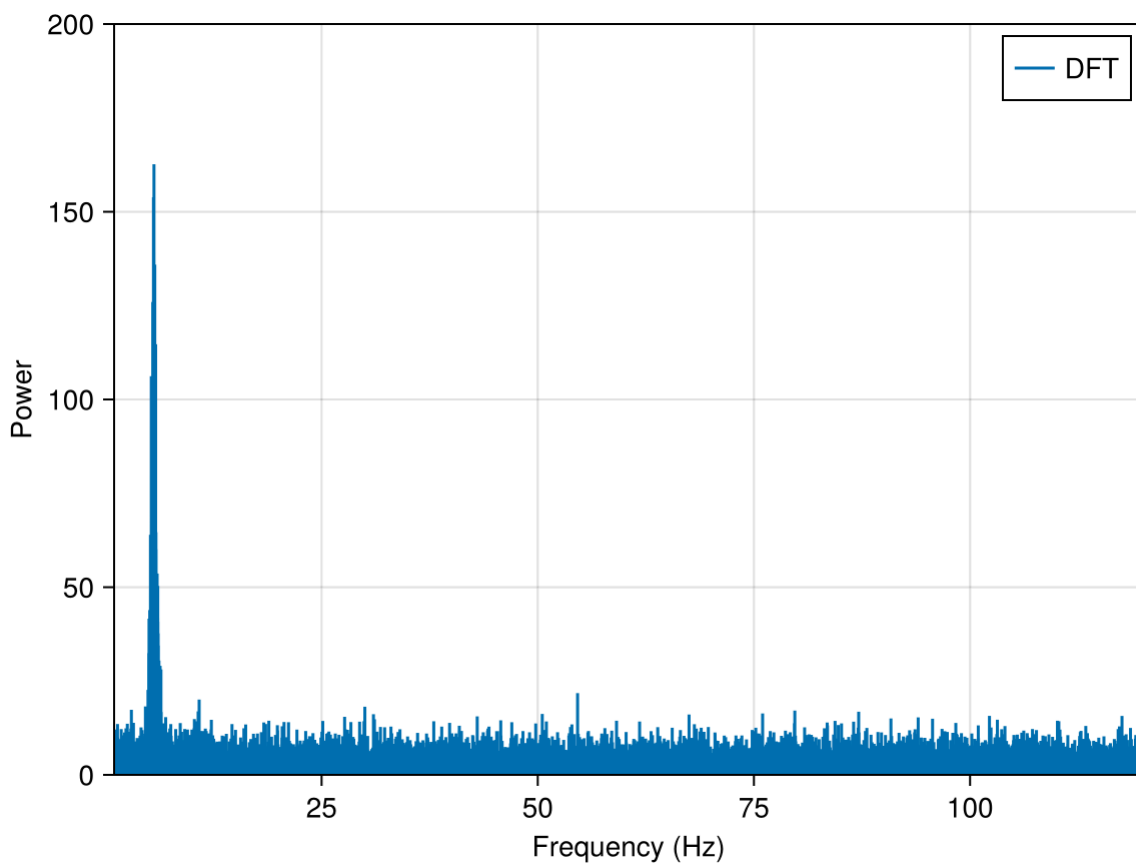


- It is difficult to see the details. With such a rapid sampling we have a very large number of points (almost 180000!). Let's try a zoom showing the first 0.5s of data.

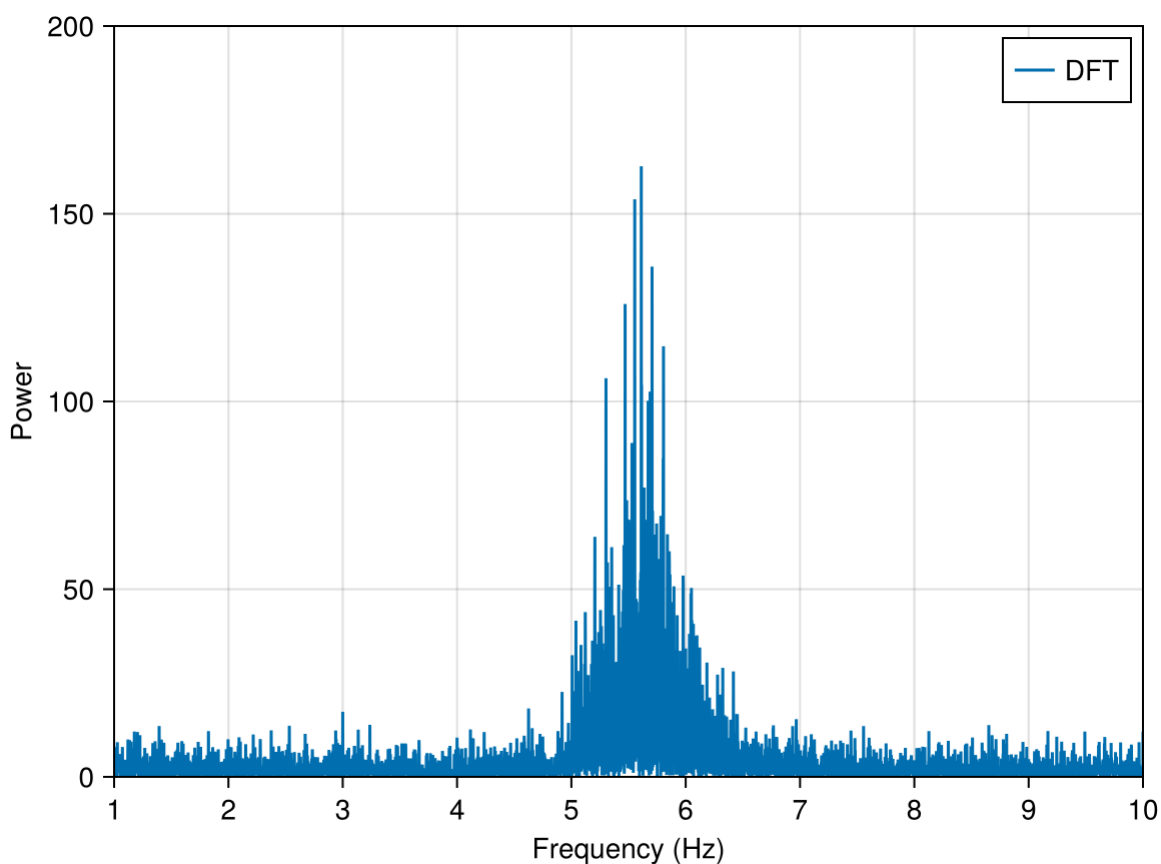
RXTE observation of GX339-4



- Let's now compute and show the periodogram



- A powerful peak is clearly visible. Again let's zoom in.



- And we can see a substructure in the peak. Indeed, it is formed by a large number of peaks with similar frequency (i.e. a QPO!). The peak centroid is close to ~ 5.6 Hz.
- We might want to test whether the identified peak can be due to noise fluctuation.

- A real analysis would ask to consider all the substructure visible in the periodogram. However, to simplify the discussion, let's pretend that the highest peak is the only one visible in the spectrum.

Maximum power: 162.65 at frequency 5.61 in a 89600 frequency bin long periodogram.

Significance of a peak

- Let's not forget that any significance has to be computed considering also the number of independent frequencies considered for the analysis. This is often the leading factor.

Significance exercise

- Let's assume to have a peak with probability 0.9997, i.e. 3σ . This is generally a rather prominent feature.
- And let's also assume our input time-series consists of 200 points, generating therefore a 100 frequency bin periodogram.
- Let's compute the final significance:- Let's compute the final significance:

FAP: 0.03%

Single-trial probability: 99.97%

Post-trial probability: 97.04%

- Quite interestingly, the final probability can justify a claim of a hint of periodicity, but nothing more: high significance periodicities need to be of very high single-trial significance.

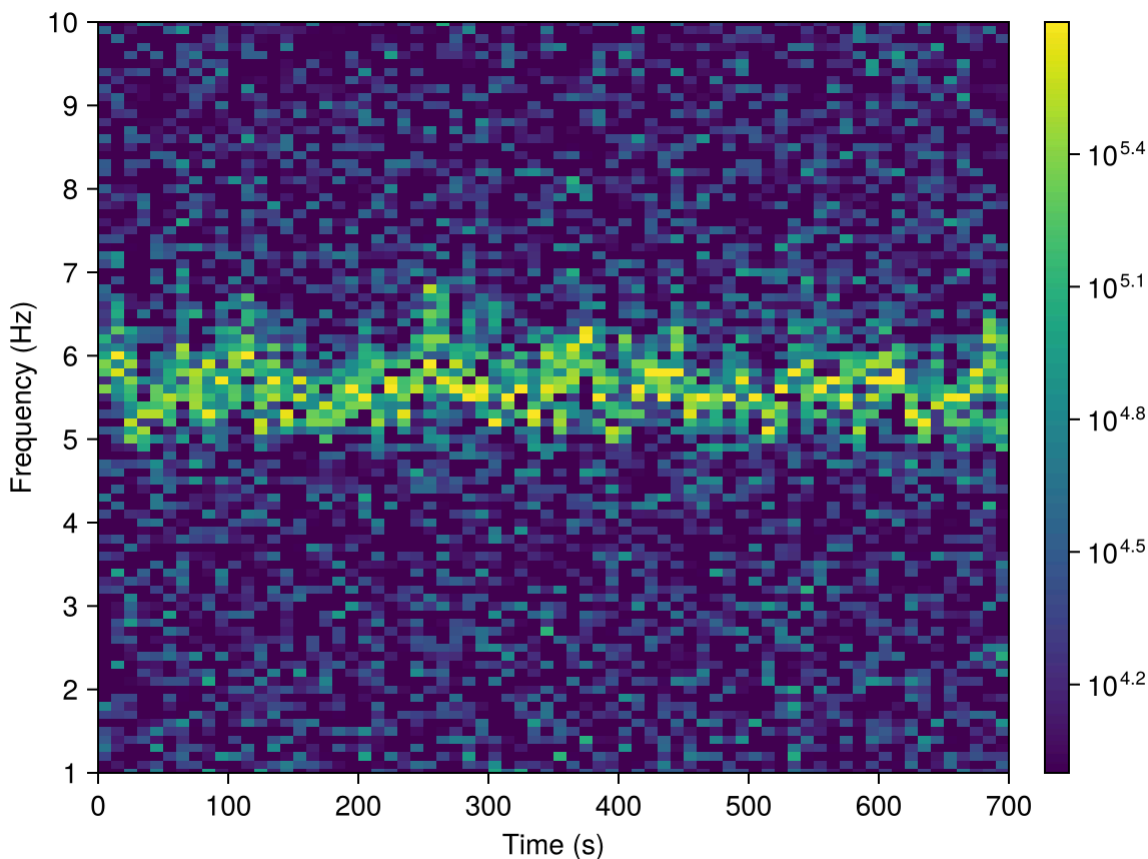
Significance of our QPO

- Coming back to our exercise, the measure power yields a very low probability to be due to noise. Yet, the number of frequencies in the periodogram is also very high.
- Nevertheless, the identified features, within the noise assumptions (white noise), are highly significant.

Single-trial probability: 100.00%

Post-trial probability: 100.00%

- The large number of peaks with similar but not identical frequency might be due to some sort of periodicity with period drifting in time.
- we can explore this possibility computing the DFT in several time-slot, in order to obtain a time-resolved DFT (more on this and other related techniques in a later lecture).



- We see that computing the periodogram in 70 10s bins the drifting of the QPO frequency is clearly visible.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- Nespoli et al. (2003) - "A transient variable 6 Hz QPO from GX 339-4"

Further Material

Papers or sites for examining more closely some of the discussed topics.

- [Bahramian & Degenaar \(2023\) - "Low-Mass X-ray Binaries"](#)
- [Tan C. \(2021\) - "High-Mass X-ray binary: Classification, Formation, and Evolution"](#)

Credits

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	<u>Science case about Sunspot number</u>	<u>Lecture about irregular sampling</u>	<u>Course Summary</u>
html	<u>Science case about Sunspot number</u>	<u>Lecture about irregular sampling</u>	<u>Course Summary</u>

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